1. **Draw a diagram of the OSI 7-layer model and label each layer with its correct name and a real-world example of a device or protocol that operates at that layer.**

Layer No.	**Layer Name**	**Real-world example of a device/Protocol**
1.	Application layer	HTTP, HTTPS, FTP
2.	Presentation layer	SSL, ASCII, TLS
3.	Session layer	NetBIOS
4.	Transport layer	TCP, UDP
5.	Network layer	Router, IP
6.	Data-link layer	Switch, Ethernet, MAC
7.	Physical layer	Ethernet cable, Fiber optic cable

2. **Match the following networking scenarios to the correct OSI layer: a) Encrypting data before sending, b) Routing packets across networks, c) Converting data to electrical signals, d) Opening a web page in your browser, e) Detecting and correcting errors in frames.**

a. Encrypting data before sending: Layer-2 Presentation layer.
b. Routing packets across networks: Layer-5 Network layer.
c. Converting data to electrical signals: Layer-6 Physical layer.
d. Opening a web page in your browser: Layer-1 Application layer.
e. Detecting and correcting errors in frames: Layer-3 Data-link layer.

3. **Pick any app you use daily (like WhatsApp, Instagram, or Zomato) and describe how at least three different OSI layers are involved when you send or receive data in that app.**

- When you send a message on WhatsApp, many OSI layers work together to deliver it successfully.

1**. Application layer:**

- In this layer, WhatsApp operates.
- It allows typing a message, attaching images or videos and sending them.

2. Transport layer:

- This layer ensures the sender's message received by the receiver.
- It manages data transmission using protocol like TCP.
- It breaks data into the segments and then reassembles them at the receiver side.

3**. Data-link layer:**

- This layer transfers data between mobile data or Wi-Fi.
- It uses MAC addresses to communicate with nearby network devices.

4. **Given this scenario: You are unable to access a website, but you can successfully ping its IP address. Identify which OSI layer(s) might be causing the problem and explain your reasoning.**

- In the given scenario, hardware-based OSI layers are not causing because ping is working as it provides IP address. However, software layers might be causing the problem.
- **Layer-1 Application layer:**
- The web server or HTTP/HTTPs service may be down, misconfigured or not responding, even though the server itself is reachable.
- **Layer-2 Presentation layer:**
- Encryption errors in HTTPs occur at this layer and can prevent secure websites from loading.
- **Layer-3 Session layer:**
- Problems can be happened between establishing or maintaining the communication session could prevent access.